

RESEARCH OVERVIEW

My research uses **Interpretability** as a lens to understand language and multimodal foundation models throughout training and inference, with a focus on **Reasoning** capabilities and **Agent** behavior. I study how these capabilities and behaviors emerge, when they fail, and how they can be evaluated, steered, and improved.

EDUCATION

- University of Maryland, College Park** College Park, MD
Ph.D. in Computer Science; Advisor: Prof. Soheil Feizi Sep 2026 - Present
- University of Maryland, College Park** College Park, MD
M.S. in Computer Science; Advisor: Prof. Tianyi Zhou; GPA: 4.00/4.00 Sep 2024 - Jun 2026
- Huazhong University of Science and Technology** Hubei, China
BEng. - Computer Science and Technology; GPA: 3.99/4.00; Ranking: Top5%; Sep 2020 - Jun 2024

SELECTED RESEARCH

Reasoning Behavior across Language and Multimodal Models

When do language and multimodal models fail to reason effectively, and what structure underlies their reasoning?

- Reasoning failures**
 - MiP-Overthinking**. Reasoning models spend substantial computation on questions with missing premises yet rarely abstain, underperforming non-reasoning models; the overthinking pattern also transfers through distillation.
 - Thinking Hard Not Smart**. Despite their strength, reasoning models fail to ration shared test-time compute: they follow presentation order and remain insensitive to question value density.
- Reasoning structure**
 - ThinkARM**. By adapting a cognitive-science theory of human problem solving, ThinkARM quantitatively distinguishes reasoning patterns across models and reveals how efficiency interventions alter reasoning structure.
 - V-REX**. Disentangling visual exploration into planning and following reveals capability gaps hidden by outcome-only evaluation.

Learning and Intervention Dynamics

When does behavior generalize during pretraining, and when do inference-time interventions succeed?

- Inference-time steering** – **ASTEER**. Steering outcomes are predictable from early hidden states, enabling efficient strength selection before full rollout.
- Pretraining dynamics** – **MoE Grokking**. Examining MoE routing pathways shows that grokking corresponds to internal structural changes despite converged training loss.

Agent Decision-Making

When should agents use tools, and why do their decisions diverge from need?

- Tool-use decision-making**
 - MetaTool**. Most tested LLMs struggle to decide whether to use tools and to select the right tools.
 - TicToc**. Agents remain poorly aligned with human preferences about when stale context requires a new tool call.
 - Knowing-Doing Gap**. Tool-need signals are internally decodable, but are not always translated into action.

SELECTED PAPERS

- [COLM 2025] Missing Premise Exacerbates Overthinking: Are Reasoning Models Losing Critical Thinking Skill? [paper] Reasoning Interpretability: Chenrui Fan*, Ming Li*, Lichao Sun, Tianyi Zhou
- [ACL 2026 Oral; SAC Highlight Award] Schoenfeld's Anatomy of Mathematical Reasoning by Language Models [paper] Reasoning Interpretability: Ming Li*, Chenrui Fan*, Yize Cheng*, Soheil Feizi, Tianyi Zhou
- [Preprint 2026] When is Your LLM Steerable? [paper] Interpretability: Chenrui Fan, Yize Cheng, Ming Li, Soheil Feizi, Tianyi Zhou
- [NeurIPS 2026] Model-Adaptive Tool Necessity Reveals the Knowing-Doing Gap in LLM Tool Use [paper] Agent Interpretability: Yize Cheng*, Chenrui Fan*, Mahdi JafariRaviz*, Keivan Rezaei, Soheil Feizi
- [Preprint 2026] Thinking Hard, Not Smart: Reasoning Models Fail to Ration Test-Time Compute Across Questions [paper] Reasoning Interpretability: Chenrui Fan*, Yize Cheng*, Ming Li, Yongyuan Liang, Tianyi Zhou, Soheil Feizi
- [ECCV 2026] V-REX: Benchmarking Exploratory Visual Reasoning via Chain-of-Questions [paper] Multimodal Reasoning: Chenrui Fan*, Yijun Liang*, Shweta Bhardwaj*, Kwesi Cobbina, Ming Li, Tianyi Zhou
- [ICLR 2026] Grokking in LLM Pretraining? Monitor Memorization-to-Generalization without Test [paper] Pretraining Interpretability: Ziyue Li, Chenrui Fan, Tianyi Zhou
- [ACL 2026 Findings] Your LLM Agents Are Temporally Blind: The Misalignment Between Tool Use Decisions and Human Time Perception [paper] Agent: Yize Cheng*, Arshia Soltani Moakhar*, Chenrui Fan*, Parsa Hosseini, Kazem Faghih, Zahra Sodagar, Wenxiao Wang, Soheil Feizi

- [NeurIPS 2025] ColorBench: Can VLMs See and Understand the Colorful World? A Comprehensive Benchmark for Color Perception, Reasoning, and Robustness [paper] [Multimodal Reasoning]: Yijun Liang, Ming Li, **Chenrui Fan**, Ziyue Li, Dang Nguyen, Kwesi Cobbina, Shweta Bhardwaj, Jiu hai Chen, Fuxiao Liu, Tianyi Zhou
- [CVPR 2024 Oral] MetaCloak: Preventing Unauthorized Subject-Driven Text-to-Image Diffusion-Based Synthesis via Meta-Learning [paper] [Privacy Multimodal]: Yixin Liu, **Chenrui Fan**, Yutong Dai, Xun Chen, Pan Zhou, Lichao Sun
- [ICLR 2024] MetaTool Benchmark: Deciding Whether to Use Tools and Which to Use [paper] [Agent]: Yue Huang, Jiawen Shi, Yuan Li, **Chenrui Fan**, Siyuan Wu, Qihui Zhang, Yixin Liu, Pan Zhou, Yao Wan, Neil Zhenqiang Gong, Lichao Sun

* Equal contribution.

INDUSTRIAL EXPERIENCE

- **Tencent, WeChat AI** Sep 2023 - Jan 2024
- *Research Intern*
 - Conducted research on personalized federated learning for mobile input methods under heterogeneous user data.

HONORS AND AWARDS

- NeurIPS 2026 Top Reviewer Oct, 2026
- ACL 2026 SAC Highlight Award Jul, 2026
- Dean’s Fellowship, University of Maryland, College Park 2026
- National Scholarship [7/348≈2.0%] Sep, 2021

TECHNICAL SKILLS

- **Programming:** Python, C/C++
- **ML Frameworks:** PyTorch, TensorFlow, Hugging Face Transformers, vLLM
- **Tools:** Git, Weights & Biases, L^AT_EX
- **Languages:** Chinese (native proficiency), English (professional proficiency)

TEACHING & ACADEMIC SERVICE

- **Teaching Assistant:** CMSC 470: Introduction to Natural Language Processing, University of Maryland, Spring 2026
- **Reviewer:** NeurIPS (Top Reviewer, 2026), ICLR, CVPR, ECCV, COLM, ACL Rolling Review (ARR)